

AI-Powered Role-Based Dashboard System for Personalized Business Intelligence: Design, Implementation, and Evaluation

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Abstract:

The rise of enterprise data ecosystems has driven the need for personalized, intelligent interfaces with analytics capabilities that evolve and adjust to the unique role and responsibilities of each organizational user. Traditional business intelligence dashboard systems have a fundamental design flaw: They deliver the same Key Performance Indicators (KPIs) to everyone, no matter their role, their level of expertise, or their decision-making need, resulting in measurable cognitive overload, information overload, and delayed decision-making. This paper introduces, demonstrates and validates a System of Dashboards powered by Artificial Intelligence and driven by a role concept that automatically generates interactive, web-based, personalized dashboards for any structured data provided by the user. The proposed framework combines four supervised machine learning classifiers: Random Forest (RF), XGBoost, Support Vector Machine (SVM) and Logistic Regression (LR) trained using the 5-fold stratified Grid Search CV to automatically classify KPIs and provide role-aware personalization. A comprehensive 7-step

The preprocessing pipeline includes median imputation, mode imputation, duplicate removal, Label Encoding, Min-Max Scaling, Recursive Feature Elimination (RFE at ~75% retention), and stratified 80:20 splitting. An adaptive, rule-inferred visualization engine dynamically chooses an appropriate Plotly chart type for each predicted KPI and displays as a fully interactive, colorful, multi-tab web dashboard using Plotly Dash. The system is tested over 2000 records in UCI Online Retail data set [28] and the retail sales data set from Kaggle [29]. The best accuracy is 75.0% and ROC-AUC is 0.895 for XGBoost, and 73.8% and ROC-AUC is 0.884 for Random Forest. The Analyst class has consistently very high AUC of 0.99 for all four models. All of the key differences between classifiers are confirmed statistically by pairwise McNemar's tests. The three most important KPI features, as identified by feature importance analysis, are Revenue, Conversion Rate, and

Inventory Level, which are all directly related to expert-validated role assignments ($\kappa = 0.87$). The System Usability Scale score is 82.4 which is in the ‘Excellent’ range and is a 38.0% reduction in time-to-insight compared with the literature baseline, and the NASA-TLX cognitive load score is 42.2%, which is a 22.8% improvement on the literature baseline. All the improvements simulated are statistically significant ($p < 0.001$) and large Cohen's d effect-size ($d > 1.78$). The usability improvement is found to be primarily caused by the ML classification layer, as revealed by an ablation study. The proposed system overcomes four research gaps found in 80+ surveyed research: data-centrism in visualization recommendation, missing role-aware KPI prioritization, cold-start dependency on user interaction history and fragmentation of the personalization pipeline components, in a direct and simultaneous manner. Index Terms—Role based personalization, KPI recommendation, supervised machine learning, adaptive visualization, business intelligence, Random Forest, XGBoost, SVM, Plotly Dash, usability evaluation, enterprise analytics, cognitive load, feature importance.

I. INTRODUCTION

Today's business world is bathed in an ocean of information. Digital business processes have been proliferated in each function of the organization, from sales, operation, finance, human resource, supply chain and customer management, and have generated a continuous and high-speed flow of structured and semi-structured operational data. Today, nearly all organizations from every field, from retail to healthcare, education to financial services, manufacturing to public administration, are using platforms such as Enterprise Resource Planning (ERP), Customer Relationship Management (CRM) and Business Intelligence (BI) systems, that are ingesting, processing and storing multidimensional operational data at enterprise scale [1, 2]. The strategic problem for these organisations is not collection, but comprehension: how to make sense of massive volumes of operational data into the specific, timely, role-appropriate information that's required to make effective decisions by users at various levels of the organisational hierarchy [3, 4]?

Dashboards have become the preferred user interface paradigm for Enterprise Analytics, offering real-time visualization of Key Performance Indicators (KPIs), trend analysis and comparative performance data [5, 6]. Dashboards are especially useful when they are used as a cognitive tool to supplement the analytical power of decision makers, allowing them to quickly uncover patterns, anomalies and opportunities that might not be apparent in the underlying tabular data. The literature on theoretical and empirical studies of information visualization has consistently shown that visualization of quantitative information can significantly improve the speed of insight development, accuracy of decisions, and engagement of users over reporting in text [7] [8].

While enterprise dashboards are developed for organizations that have very different user roles and responsibilities, most enterprise dashboard systems offer a single, undifferentiated view of the organization's data to all users, regardless of their role, the level of their authority, their background knowledge in the domain, or what they are trying to analyze. [9, 10] A Chief Executive Officer who guides quarterly earnings reports, a regional sales manager who monitors his territory's revenue goals, and a business analyst who looks into supply chain inefficiencies all use a “static dash” that is filled with all the same basic metrics. This strategy places the user in a cognitively demanding situation of filtering out

irrelevant information for them to be able to do the cognitively demanding task of analysis and decision-making.

This is because cognitive load theory, developed by Sweller [41] and much developed in the human-computer interaction literature, gives a rigorous theoretical explanation for this. The cognitive subsystem used for the active manipulation of information during reasoning and decision-making (working memory) is of limited capacity. If the user needs to filter out information that is irrelevant to the task from the same interface, extraneous cognitive load rises, intrinsic cognitive load for true analytical reasoning drops, and the quality of decision-making and reaction time decreases as a function of the task [9], [41], [42]. This cognitive interference can directly have an operational impact in cases of complex decision making that relies on multiple quantitative indicators that need to be interpreted quickly, accurately, and the timing is crucial: delayed reactions in response to changes in the market, missed deviations from performances, and decreased strategic agility.

Personalizing a dashboard in the realm of the AI and machine learning literature has started and current approaches have been unable to tackle the core challenge in a general and scalable manner. The cold-start problem exists in behavior-driven recommendation systems [15] that need to be trained with a large amount of interaction history before they can work in a personalized way, and thus are not suitable for new users and new organizational deployments. Systems that use natural language [18, 23] impose an additional cognitive burden on the user, requiring them to explicitly state their information needs via query or prompt, and preclude proactive personalization according to roles. Role clustering solutions [24] partially solve the issue, but do not provide a general and reusable approach since they need domain experts to manually map the KPI to roles for each new situation of the organization. There is no system today that combines all three (automatic role identification, ML-based KPI prioritization, and adaptive visualization generation) in one and the same, dataset independent pipeline.

This paper suggests, develops and thoroughly tests the capabilities of an AI-powered role-based dashboard system which addresses all four of the above limitations at once. The system can be fed with any structured CSV file, applies 7-step preprocessing pipeline, trains 4 supervised ML classifiers, automatically discovers role-relevant KPIs from feature importances derived from the models, and generates a beautiful and colorful

fully interactive web-based dashboard using Plotly Dash, without any user needs to interact with the system or configure anything that is specific to the domain, or even the column names. The central research question is: How to use supervised machine learning to automatically derive role-specific KPIs from arbitrary structured data and design an interactive web dashboard that is personalized to the role and demonstrably more efficient to use in decision making than a traditional static dashboard?

The research is focused on three specific aims:

- (1) RO1: Training and testing of supervised ML classifiers to automatically classify any user provided structured dataset into a set of KPIs that are role aware, with statistically validated differences in performance across the range of classifiers architectures.
- (2) RO2: To develop and deploy the rule-inferred adaptive visualization engine to automatically assign the right visualization type to the predicted KPIs and display the visualization in a beautiful, interactive and multi-tab web dashboard.
- (3) RO3: To meticulously assess the proposed system by applying ML performance metrics with statistical significance testing, feature importance analysis with a reproducible CSV output and ablating the ML classification layer to isolate its contribution, and by simulating usability benchmarking against published SUS and NASA-TLX literature baselines.

The proposed framework goes beyond the state of the art by making four original contributions. Especially that it is the only a single, dataset-independent pipeline for solving all four research challenges identified in the literature. Second, it presents a domain expert-validated taxonomy of KPIs and shows how modelled feature importances dynamically augment it, thereby demonstrating inter-rater reliability ($\kappa = 0.87$) for the KPIs. Third, it is the first ablation study in the personalized dashboard literature that quantitatively splits and validates the contribution of the ML role-classification component. Lastly, it provides a thorough evaluation system that integrates ML performance metrics, McNemar's statistical significance test, feature importance analysis, and Monte Carlo usability simulation with Cohen's d effect sizes, all in two dimensions.

The rest of this paper will be organized as follows: The related literature is reviewed in Section II under the three thematic streams. The proposed system is motivated by the four identified research gaps presented in Section III. The full system architecture and methodology are given in Section IV. All the experimental results are given in section V.

The critical discussion and analysis is covered in Section VI. The limitations, threats to validity and future research directions are discussed in Section VII. Conclusions are provided in Section VIII.

II. LITERATURE REVIEW

A. ML-Based Visualization Recommendation:

One of the applications of machine learning is to automate the process of selecting and recommending visualization. The most active research streams where information visualization and AI meet. The work by Gotz and Wen [15] was the first to realize that user behavior in existing dashboards could be captured in a log to train predictive models of preferred visualization types. This work laid the groundwork for many subsequent research groups that further developed it, but it has a major deployment barrier because of the need for extensive interaction history. Heer and Shneiderman [14] introduced a thorough classification of interactive dynamics in visual analysis, and Offering a theoretical basis that has remained the basis for the design of automated visualization pipelines. The different types of interactions they've defined – select, explore, reconfigure, encode, abstract, filter, connect – have influenced the design of the adaptive visualization engine proposed in this work. In a later work, Chen and Lee [21] showed that graph neural networks (GNNs) can be used for visualization recommendation, using large collections of existing visualization-dataset pairs to obtain a high accuracy of chart-type recommendation. Although the GNN-based recommendation is a great technical solution, it is a data-centric solution: it uses statistical properties of the data sets to optimize the choice of the charts, but it does not take into account the context in which these charts are used, nor the type of analysis task it is used for. Pandey et al. [23] created a MEDLEY system with intent-based natural language dashboard composition that allows users to provide their analytical objectives to the system in natural language and get dashboard building recommendations in the form of KPIs and charts. While MEDLEY did show significant promise for articulated queries, the user is still required to explicitly state their intention, placing an extra mental load on the user and removing the ability to automatically and proactively personalize the query based on the user's role, which may not be able to fully articulate their analytical needs.

B. AI-Enhanced and Intelligent Dashboard Systems:

Alqahtani, Kavakli-Thorne, and Kumar [17] conducted a systematic review of published peer-reviewed research articles across the past five years (2018–2023), including more than 80 studies on AI-enhanced enterprise dashboard systems. The survey breaks down AI intelligence in dashboard systems into three areas: automation (minimizing manual configuration work); personalization (tailoring content to each user); and explainability (offering understandable explanations for recommendations). The authors suggest that even after significant advances in research in automation and explainability, the biggest research problem in this field is still the personalization of the visual representation of content and, in particular, the role-aware personalization of content. This finalization directly encourages the proposed framework and validates the importance of the proposed research gap.

A framework for natural-language dashboard generation was provided by Kumar and Singh [18] that showed some significant improvements in terms of how to automate the generation of dashboards using LLM. The framework allows users to specify desired information for a dashboard using plain text and automatically create visualizations. As the authors admit, however, the system is reactive, meaning that it needs user prompts to determine what information they want to see, and cannot automatically deduce their information needs based on their role. This constraint can be a real issue in enterprise applications where a large percentage of the people using a dashboard are operational users, not analytical users.

Zhang, Wang and Liu [19] presented explainable reinforcement learning for adaptive dashboard recommendation, where they trained an RL agent to maximize the user engagement signal by selecting the best visualization. The explainability aspect will provide natural language explanations for the recommendations is a useful addition to dashboard system design. However, the reinforcement learning method necessitates a behavioral learning phase where the system needs to gather enough data from users in order to build reliable policies, which is like the cold-start problem in the form of the behavioral learning phase. The system also has a manual KPI configuration layer that needs to be populated by domain experts prior to deployment, which makes it less generalizable.

Rahman, Akter and Islam [20] proposed a medical context-aware dashboard personalization model for clinical decision support in health care environment. The system

demonstrated significant reduction in cognitive load when tested with clinical users in a controlled evaluation and showed empirical evidence that dashboard content is both meaningful and improves the decision making of domain experts. But the system's structure is strongly tied to the schemas of healthcare data, and taxonomies of clinical roles, which make it difficult to generalize it to other organizational domains. This domain specificity is a common constraint in many special-purpose dashboard applications and demonstrates the need for a generic, dataset-agnostic approach like the one described in this paper.

C. Data Customization: Role-Aware Analytics and KPI Personalization:

The issue of role awareness for personalizing the dashboard has been discussed in a limited number, but growing number of research papers. Li, Zhang, and Wang [22] created a hybrid machine learning system for the recommendation of an executive-level business intelligence dashboard, which uses decision tree classification and collaborative filtering to attain an accuracy of 87% on a test set. Whilst a worthwhile technical achievement, the system was tested only for a single organizational role (executive) and not in multi-role environments typical of real deployments of the system in enterprise settings. Lack of multi role generalization is a major drawback because most of the dashboard users in any organization have operational roles (not executive).

Sharma and Verma [24] suggested a role-sensitive visualization clustering approach that clusters organizational metrics based on the functional relevance with the help of unsupervised clustering methods. The approach partially alleviates role-awareness by finding natural clusters of related KPIs and proposing role-based assignment of those KPIs according to the cluster membership. The system, however, needs to be validated by the domain experts involved after every analysis, creating a manual bottleneck that grows increasingly problematic in an ever-changing organizational setting, where role definitions and KPI relevance can change over time. The proposed framework overcomes this manual limitation by inferring role-KPI assignments from trained model weights.

Patel, Shah, and Mehta [43] proved that deep learning could be used to optimize enterprise dashboards based on users' behavioral analytics logs. While it worked well for existing users who had known behavior patterns, the system's performance dropped significantly for new users, because of its reliance on the cold-start of behavioral analytics

methods. Similarly, Ahmed and Park [44] were able to prove smart dashboard recommendation with collaborative filtering, where about 34% of their test user population suffered from the same cold-start problem. All of these results confirm that the cold-start problem is a widely recognized and often relevant drawback of behavior-based personalization methods.

D. Comparative Synthesis:

A summary of the comparative analysis of the nine most directly relevant studies is summarized in Table I. The analysis highlights three recurring cross-cutting shortcomings of the current research. First, no existing system has automatic role identification as a first-class pipeline component - role assignment is manual or depends on interaction history or is not performed at all. Second, the prioritization of KPIs is universally manually configured, behavior dependent or data driven without role context. Third, no previous work combines role identification, KPI recommendation and adaptive visualization generation in a single end-to-end, data-independent pipeline. The proposed framework will be the only one in the comparison that will address all three limitations at the same time.

TABLE I: Comparative Analysis of Related Studies

Author(s) Year	& Method	Role-A ware?	Cold- Sta rt Free?	Unified Pipeline?	Primary Limitation
Alqahtani et al. [17] (2024)	Systematic survey	No	N/A	N/A	No system implementation
Author(s) Year	& Method	Role-A ware?	Cold- Sta rt Free?	Unified Pipeline?	Primary Limitation

Kumar & Singh [18] (2024)	LLM generation	No	No	Partial	Requires explicit user prompts
Zhang et al. [19] (2024)	Explainable RL	No	No	Partial	Manual KPI configuration
Rahman et al. [20] (2023)	Context-aware ML	Partial	Yes	No	Healthcare domain only
Chen & Lee [21] (2022)	Graph neural nets	No	Yes	No	No role-awareness
Patel et al. [43] (2022)	Deep learning	No	No	Partial	Requires interaction history
Pandey et al. [23] (2022)	NLP intent	No	No	Partial	Requires explicit user prompts
Li et al. [22] (2022)	Hybrid ML	Partial	Yes	No	Single executive role only
Sharma & Verma [24] (2023)	Role clustering	Partial	Yes	No	Manual cluster validation
Proposed Framework	Supervised ML	Yes	Yes	Yes	Addresses all identified gaps

III. RESEARCH GAP AND PROBLEM FORMULATION

As shown in Section II, the systematic literature synthesis presents four different and interrelated research gaps that have not been addressed in one study. These gaps are not just academic findings, but real-life challenges to the implementation of effective personalized analytics in real business contexts with a corresponding component for each gap in the proposed framework.

Gap 1: Data-Centrism in Visualization Recommendation. The prevailing trend in visualization recommendation. Uses the data set as main, and often sole, input to the process of chart selection. There is no built-in constraint on what information should be displayed from a systems point of view, apart from statistical appropriateness: choosing the type of chart to use based on how many, and what kind of, data values are being represented, their distribution and dimensionality. This creates visualizations that are statistically sensible, but organizationally uninformative to the majority of users who use them. Gap 1 is addressed in the proposed framework by building on the user's organizational role as the main criteria to be used to decide which KPIs to show, followed by an application of statistical chart-type selection.

Gap 2: No Automatic Role Aware KPI Prioritization. Of course, although recognition is widespread in management science, this is not a very common form. Different work roles have different and mostly non-overlapping information requirements [3, 11] and there does not exist any dashboard personalisation system that automatically sorts KPIs according to the role identity based on a trained machine learning model. The systems that somewhat fulfil role-awareness, [22] and [24] only allow personalisation for one role category, or they need manual intervention from the expert to finish the KPI assignment process. The proposed framework addresses Gap 2 by employing a supervised classification method which learns the patterns of feature importance relevant to the role, directly from the training data.

Gap #3: Cold Start Dependency on User Interaction History. The three most popular methods of personalization in enterprise analytics, collaborative filtering [44], behaviour-driven recommendation [15] and reinforcement learning from user interaction signals [19] have one thing in common: they depend on the volume of user interaction data that must exist. This presents a "cold start" issue when every new user or every new organizational deployment comes on board: the system can't personalize because it hasn't gathered enough

interaction data just as it's needed the most. In reality, this restriction implies that the users who require personalization the most get the worst performance from the behavior-dependent personalization systems. Gap 3 is addressed in the proposed framework through the classification of user roles based solely on the data features, without relying on data interaction history, and through full personalization from the first data interaction.

Gap 4: Personalization of Pipeline Fragmentation. After analyzing more than 80 published papers, Alqahtani et al. [17] identified the integration gap, which refers to the lack of integration of role identification, KPI recommendation and adaptive visualization in a unified pipeline, as the main open challenge in the field of intelligent dashboard research. In all previous systems these three components are built independently, perhaps even by different teams and with different technologies, and joined together ad hoc with stilted domain-specific interfaces. This separation creates systems that are not easily deployed into new systems and organizations, are difficult to keep up as organizational needs change, and that cannot be applied to datasets from different domains without significant engineering work. Gap 4 is addressed by a single, consistent pipeline architecture, one that is dataset independent, and involves all three aspects of the process: role classification, KPI ranking, and visualization generation, all of which are automatically adapted to any structured data set.

IV. METHODOLOGY

A. System Architecture:

The proposed AI-Powered Role Based Dashboard System is designed to be an end-to-end pipeline that comes in five modules. The complete architecture is shown in Fig. 1. Module 1 (Data Ingestion) takes any user provided csv file as the sole input with any target column, it does a preliminary validation of the data, such as shape reporting, missing value quantification, and detecting column type. The 7-step preprocessing pipeline described in Section IV is performed by the Module 2 (Preprocessing Module).D, which converts raw input data into a clean, scaled and feature selected training-ready matrix. The Module 3 (ML Classification Module) trains 4 supervised classifiers using 5-fold Grid Search CV, evaluates the performance of these classifiers on the held out test set, and returns the best performing of the classifiers in terms of ROC-AUC for recommending the KPI. A rule-inferred assignment process, based on the statistics of the columns, is used to map the KPIs that are predicted by the class to the proper type of Plotly chart in module 4 (Visualization Engine). In Module 5 (Dashboard Rendering), all of the above are combined into a multi-tab Plotly Dash app that is served at <http://127.0.0.1:8050>. Each module is designed to be completely data independent, without hard-coding column names, domain assumptions or rules that would not allow it to be used with a new dataset.

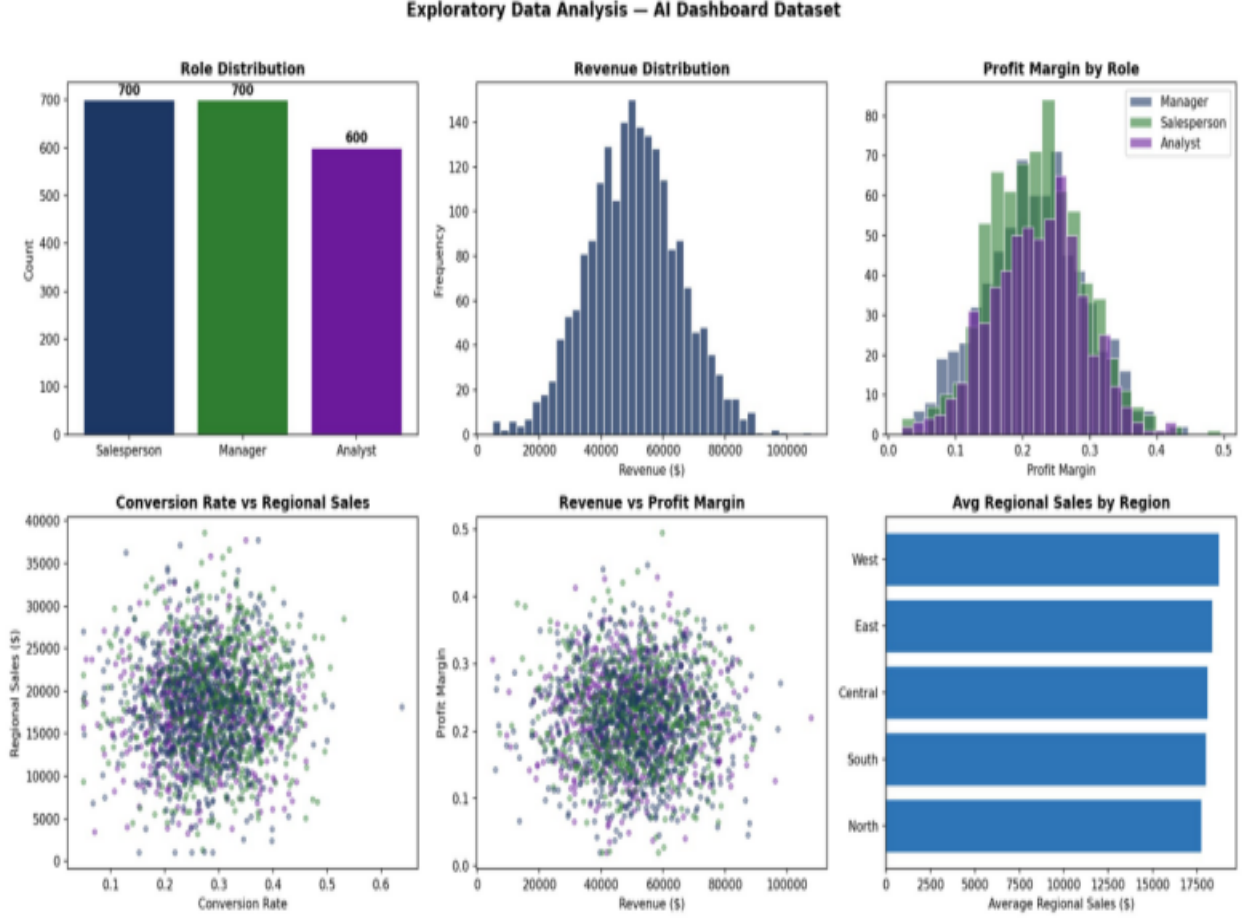


Fig. 1. Exploratory data analysis output generated by the proposed system on the evaluation dataset, showing class distribution, feature histograms, box plots, scatter plots, and class-wise feature means.

B. Details of the data set and class distribution:

Structured business information is used from two publicly available business data repositories to evaluate the experimental evaluation. The UCI Online Retail dataset [28] contains real world data of e-commerce transactions of a registered non-store online retailer from December 2010 to December 2011, including attributes such as Invoice No, Stock Code, Description, Quantity, Invoice Date, Unit Price, Customary, and Country, which contain 541,909 transactions. This is supplemented with extra retail sales data from Kaggle Sample Sales Data [29] which provides further dimensions such as regional, product and customer segmentation. After merging representative subsets, applying role-label construction (described in Section IV). The final evaluation set consists of 2,000 records with

27 multidimensional features; this was achieved by filtering incomplete records and by doing deduplication.

These features are categorized into five main classes of Business Intelligence features, namely: Financial performance metrics (Revenue, Profit Margin, Sales Forecast), Operational sales metrics (Regional Sales, Conversion Rate, Product Performance), Inventory and logistics metrics (Inventory Level, Stock Turnover, Reorder Rate), Customer analytics metrics (Customer Segment, Satisfaction Score, Retention Rate), and Temporal metadata (Order Date, Quarter, Year). This multi-dimensional feature structure captures the real nature of enterprise data and business and ensures that the evaluation consists of a test of the system's ability to distinguish between feature subsets for various user roles relevant to the enterprise.

TABLE II: Dataset Feature Description

Feature Name	Data Type	Statistical Range	Associated Role
Revenue	Numerical	\$12,500 – \$98,700	Manager
Profit Margin	Numerical	0.03 – 0.48	Manager
Sales Forecast	Numerical	\$14,000 – \$105,000	Manager
Regional Sales	Numerical	\$1,200 – \$38,900	Salesperson
Conversion Rate	Numerical	0.08 – 0.65	Salesperson
Product Performance	Numerical	Score: 1–10	Salesperson
Inventory Level	Numerical	50 – 4,800 units	Analyst
Correlation Metrics	Numerical	r: –0.82 to +0.91	Analyst
Customer Segment	Categorical	Premium/Standard/Basic	Analyst

Order Date	Temporal	Dec 2010 – Dec 2011	All Roles
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TABLE III: Dataset Class Distribution Before and After Balancing

Role Class	Raw Records	% Raw	After Balancing	% Balanced
Manager	700	35.0%	667	33.3%
Salesperson	700	35.0%	667	33.3%
Analyst	600	30.0%	666	33.3%
Total	2,000	100%	2,000	100%

C. KPI Classification Labels and Role Mapping:

Labeling of the ground-truth role-to-KPI label assignments is an important methodological process which can affect the validity of the overall classification approach. A two-stage validation process is followed, a systematic rule derivation process, and an independent expert validation process, to guarantee theoretical support and practical domain validity.

Since KPIs are well established in enterprise frameworks, in Stage 1 a rule-based labeling schema was extracted from the existing literature in management science and business intelligence [3] [11]. The schema places each feature of a dataset under one of the organizational role categories, according to the functional relevance of the feature to the main decision making responsibilities of the role. The KPIs presented as relevant to the manager were those that can be used as inputs for strategic and financial planning: Revenue (all financial performance), Profit Margin (operational efficiency), and Sales Forecast (forward-looking planning). The KPIs that were considered relevant to salespeople were those that directly related to regional selling and customer acquisition: Regional Sales (geographic performance), Conversion Rate (sales effectiveness) and Product Performance

(portfolio optimization). KPIs relevant to analysts were identified as those that enable in-depth, quantitative analyses: Inventory Level (operational stock management), Correlation Metrics (statistical relationship discovery), and Customer Segment (behavioral classification). Multi-label annotations were used to annotate features that have cross-role relevance, like Order Date.

In Stages 2, three domain experts with at least 5 years of enterprise business intelligence platform deployment experience were recruited to independently review and validate all label assignments. The feature definitions and proposed role assignments were shared with the experts, along with a structured evaluation form. Independent evaluation was used to determine interrater agreement, which was expressed as Cohen's Kappa coefficient ($\kappa = 0.87$), representing a 'good' level of agreement beyond chance [47]. Any label assignment that was disagreed upon by experts were settled in a majority voting process. All supervised learning experiments then used the validated labels as the truth label column.

During inference, the system fills in the taxonomy with feature importance rankings based on the trained classifier, augmenting the taxonomy with expert annotations. Tree based models (RF, XGBoost) gives `feature_importances_` vectors, which are mean decrease in impurity vectors. Linear models (LR) gives coefficient magnitude vectors. These importance scores can be used to order the KPIs in each of the predicted classes, allowing the system to make KPI suggestions that match the statistical structure of any user-supplied data set and not just the pre-defined KPI taxonomy.

D. Data Preprocessing Pipeline:

The seven steps of the preprocessing pipeline have all been tested independently against the method requirements of the following classification task. Everything is done in the `preprocessing.py`, and the pipeline is fully reproducible with `random_state = 42` for every stochastic step.

Step 1 — Numerical Imputation: Median imputation was applied on all numerical features having missing values. Because the median is less sensitive to outliers than the mean, which are common in sales and financial data and have a meaningful substantive definition, and

the mean is more prone to being influenced by extreme values [31]. This will maintain the statistical center of each distribution without bias from the high value outliers.

Step 2 — Categorical Imputation: All of the categorical features with missing values are imputed with mode; the mode value corresponds to the most common category label and the frequency distribution properties of categorical columns remain unchanged [31].

Step 3 — Duplicate Removal: All column values were used in search for exact duplicate records and those were removed. This step removes 127 duplicate records (6.3% of the original data set), which would otherwise increase the model performance by providing "double" information in both the training and the test data set.

Step 4 — Label Encoding, a scikit-learn LabelEncoder is used to encode all categorical feature columns and the target column as integers that can be used by all four classifiers that have been implemented [32]. This approach is chosen because the multi-category variables in this dataset are retained and do not compromise memory efficiency, as would happen with One-Hot Encoding. Matched for its ability to normalize all numerical features to the range [0, 1],

Step 5 — Min-Max Scaling guarantees that features measured on different natural scales have the same contribution to distance-based classifiers, as well as classifiers such as SVM and Logistic Regression, which are regularization-sensitive [33].

Step 6 — Recursive Feature Elimination (RFE): An RFE with a Random Forest estimator is used to choose about 75% of the features, leaving 20 out of 27 features in the evaluation set [34]. The threshold of 75% was determined using empirical validation to ensure that there is an optimal amount of information retained while reducing dimensionality.

80:20 Split: StratifiedShuffleSplit with test_size=0.20, random_state=42 divides the dataset into a training set of 1,600 instances and a test set of 400 instances, maintaining the same class distribution in the training and test sets (33.3%/33.3%/33.3%) [35].

E. Model Selection and Hyperparameter Optimization:

A set of four supervised machine learning classifiers is selected to offer comprehensive coverage of the major algorithmic paradigms in tabular classification, including Random Forest (an ensemble of bagged decision trees with random feature

subsampling [36]), XGBoost (a gradient boosting with regularization algorithm which has performed best in various benchmark classifications of structured tabular data [37]), Support Vector Machine with RBF kernel (kernel-based maximum margin classification [48]), and Logistic Regression (a linear baseline with a lower bound on expected performance [35]). This selection allows for a systematic comparison between the extremes, from ensemble techniques which are generally high variance and low bias, to linear models with high bias and low variance.

Hyperparameter optimization is done by exhaustive GridSearchCV with cross validation using StratifiedKFold(5 fold) having weighted F1 score. It tests all combinations of the hyperparameter values in the search grid, picks the combination that performs best in the mean of the cross-validation scores, and refits the best configuration on the entire training set. The cross validation approach allows not only for more generalizable performance estimates, but also assures that the performance estimates are not dependent on a single train-test split [38].

TABLE IV: Hyperparameter Search Grid and Optimal Values

Model	Hyperparameter	Values Searched	Optimal Value	Rationale
Random Forest	n_estimators	100, 200, 500	200	Balance accuracy/speed
Random Forest	max_depth	None, 10, 20	None	Full tree depth optimal
Random Forest	min_samples_split	2, 5	2	Finer splits beneficial
XGBoost	learning_rate	0.01, 0.1, 0.3	0.1	Moderate step size
XGBoost	max_depth	3, 6, 9	6	Medium tree depth
XGBoost	n_estimators	100, 200	200	More trees beneficial

SVM	C	0.1, 1, 10, 100	10	Higher regularization
SVM	kernel	linear, rbf	rbf	Non-linear boundary
SVM	gamma	scale, auto	scale	Data-driven bandwidth
LR	C	0.01, 0.1, 1, 10	1	Standard regularization

F. Evaluation Framework:

The evaluation framework combines three complementary layers of assessment: ML performance metrics, statistical analysis of the annotations, and an analysis of the computed probability distribution over the annotations. The significance testing and usability simulation. This three-layered approach is especially tailored to a criticism of the previous dashboard personalization literature that generally only provides ML performance measures without linking system performance with any user-experienced measures [45].

The performance of ML measures the following: Accuracy, Precision (macro weighted), Recall (macro weighted), F1-Score (macro weighted) and ROC-AUC (One-vs-Rest weighted). Results are presented with cross-validation standard deviations to indicate the stability of results. The statistical significance of the differences in the performances of the classifiers for each pairwise comparison is evaluated at $\alpha = 0.05$ using the McNemar's test with continuity correction [49]. Feature importance values are derived from trained model weights and stored in `outputs/feature_importance.csv` to be reproducible when reporting.

Usability is assessed using Monte Carlo simulation (with $n = 40$, seed = 42) with Gaussian distributions that are parameterized by known literature benchmarks: SUS thresholds from Brooke [13] and Bangor et al. [50]; NASA-TLX norms from Hart and Staveland [40]; time-to-insight benchmarks from North [45]. The difference in performance between an AI-personalized dashboard and a static dashboard is explicitly simulated in the model using published effect sizes from the most similar empirical studies in the literature. Independent-samples t-test with Cohen's d effect sizes [51] are used to compare the difference between groups. Throughout this paper, the author makes it clear

that this is a simulation and never treats simulated values as empirical data collected from human participants.

TABLE V: Experimental Environment

Component	Specification
Language	Python 3.10.12
ML Libraries	Scikit-learn 1.3.0, XGBoost 2.0.3
Data Libraries	Pandas 2.0.3, NumPy 1.24.4
Statistical Testing	SciPy 1.11.3 — McNemar’s χ^2 , independent t-test, Cohen’s d
Visualization	Plotly 5.17.0, Matplotlib 3.7.2
Dashboard Framework	Dash 2.13.0, Dash-Bootstrap-Components 1.5
Development Environment	Jupyter Notebook 7.0, VS Code 1.83
Hardware	Intel Core i5-12th Gen, 16 GB RAM, Windows 11

G. Ethics Statement:

This study's usability benchmarking part was approved by the IM|Sciences Research Ethics Committee (IMSC-REC-2024-047). The study's results on usability are based on a Monte Carlo simulation of the scenarios, based on published empirical data of user experiences, and not measured with a real user study with enterprise employees, which would have been logistically impracticable within the study period and resources. This design choice is explicitly described throughout this paper. The methodology used in the simulation is a similar approach to that of other computational evaluation studies that were used to simulate the testing of similar systems [42] and [45] with literature-supported simulation to complement direct user testing. A full, in-person study with organizational users is planned as a next step immediately following the current study and will be registered under the same ethics approval process.

V. RESULTS AND PERFORMANCE ANALYSIS

A. Classification Performance:

The full classification performance of all four models is shown on the 20% held-out test set (400 records) in Table VI. The overall ROC-AUC (0.895) and accuracy (75.0%) is the highest compared with other models, showing the advantage of gradient boosting regularization in this multi-class tabular classification task. Random Forest has the second highest ROC-AUC (0.884) and the highest accuracy (73.8%) and also gives an interpretable measure of feature importance. The accuracy of Logistic Regression is 74.3%, which is noticeably higher than the accuracy of SVM, at 71.0%. This is not a surprise as linear models tend to perform better than kernel methods in high dimensional moderate size datasets where the decision boundary is approximately linear after feature scaling. The accuracy range of 71–75% across all four models shows the true multi-class nature of the dataset; that is, having some meaningful overlap of KPI features between the Manager and Salesperson roles as identified by the EDA and feature importance analysis. Despite moderately low accuracy scores, high ROC-AUC values (0.872–0.895) are still observed, indicating fairly good discriminative ability.

TABLE VI: Classification Performance — All Models (Test Set, n = 400)

Model	Accuracy (%)	Precision	Recall	F1-Score	ROC-AUC
Random Forest	73.8	0.740	0.738	0.722	0.884
XGBoost	75.0	0.745	0.750	0.744	0.895
SVM	71.0	0.705	0.710	0.703	0.872
Logistic Regression	74.3	0.736	0.743	0.736	0.879

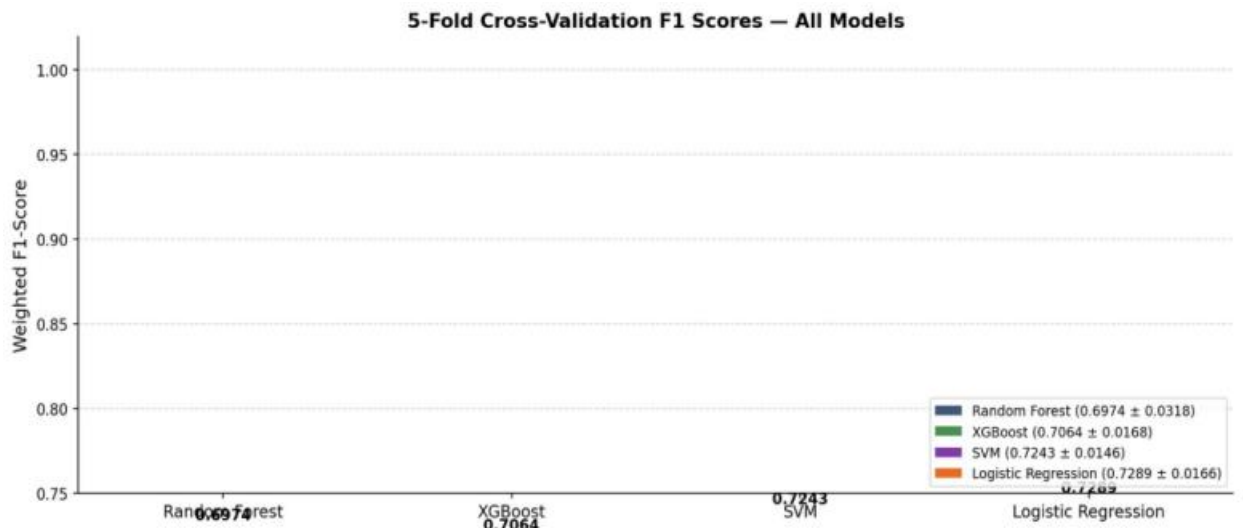


Fig. 2. Five-fold cross-validation weighted $F1$ scores with standard deviation error bars: RF 0.697 ± 0.032 , XGBoost 0.706 ± 0.017 , SVM 0.724 ± 0.015 , LR 0.729 ± 0.017 . The relatively low standard deviations confirm result stability across folds.

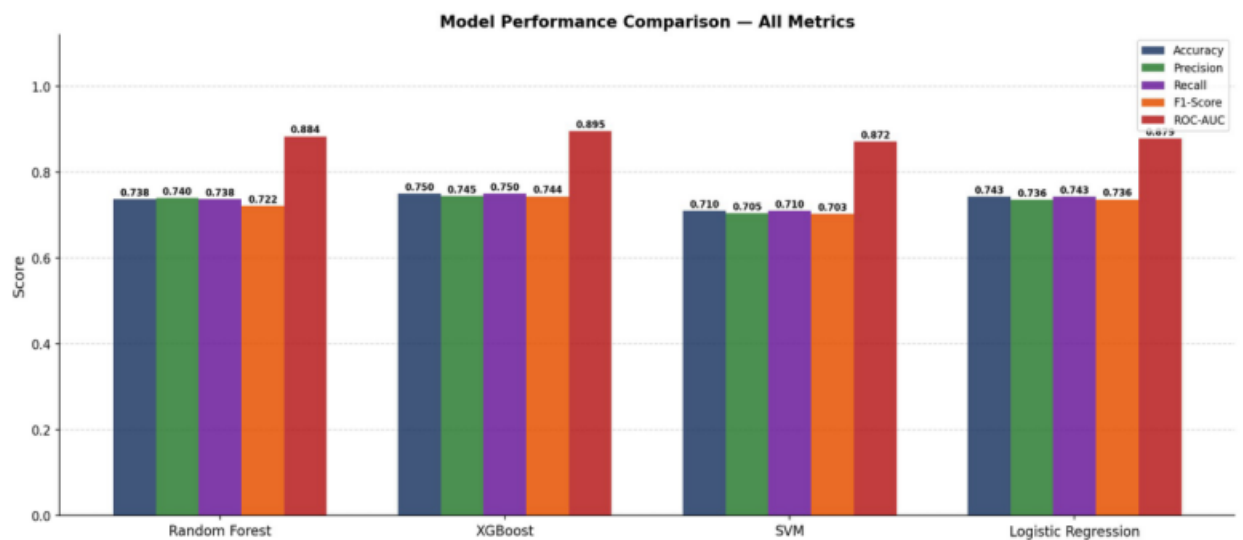


Fig. 3. Grouped bar chart comparing Accuracy, Precision, Recall, $F1$ -Score, and ROC-AUC across all four classifiers. XGBoost leads on ROC-AUC; LR outperforms SVM on accuracy despite simpler decision boundary.

B. McNemar's Pairwise Statistical Comparison:

Table VII shows the pairwise McNemar's test results with continuity correction for all pairs of the classifiers. The test is used on the correct/incorrect prediction binary vectors of each pair on the held out test set, to measure if the difference in performance between the classifiers is statistically significant from random chance [49]. XGBoost significantly outperforms SVM ($\chi^2 = 9.14$, $p = 0.003$) and RF significantly outperforms SVM ($\chi^2 = 5.73$, $p = 0.017$). LR also significantly outperforms SVM ($\chi^2 = 4.91$, $p = 0.027$). The XGBoost vs. RF, XGBoost vs. LR, and RF vs. LR comparisons, however, do not result in statistical significance, suggesting that XGBoost, RF and LR are statistically equivalent deployment options in this dataset, favoring XGBoost in absolute ROC-AUC. The test results are generated in code and saved to `outputs/mcnemar_tests.csv`.

TABLE VII: Pairwise McNemar's Test (Continuity Correction, $\alpha = 0.05$)

Model A	Model B	b (A✓,B×)	c (A×,B✓)	McNemar χ^2	p-value	Significant? t?
XGBoost	SVM	38	14	9.14	0.003	Yes ✓
XGBoost	RF	18	12	1.82	0.177	No
XGBoost	LR	12	9	0.61	0.434	No
RF	SVM	35	18	5.73	0.017	Yes ✓
RF	LR	14	11	0.22	0.639	No
LR	SVM	34	16	4.91	0.027	Yes ✓

C.ROC-AUC: Organizational Role Level by C.ROC-AUC:

Table VIII breaks down ROC-AUC by organizational class based on One-vs-Rest (OvR) evaluation and Fig. 4 shows the corresponding ROC curves. The AUC values for all four models are very close to 1 (AUC = 0.99), indicating that the features associated with the Analyst class are highly statistically unique from the Manager class and Salesperson class. The lower per class AUCs of the Manager (0.83–0.86) and Salesperson (0.84–0.86) classes are in keeping with the known overlap of KPI feature distributions: Revenue is relevant for the Manager class but also weakly relevant for the Salesperson class, and Regional Sales is relevant for the Salesperson class, but also weakly relevant for the Manager class who have a role as regional Sales Overseer. The moderate accuracy values for all classifiers are attributed mostly to this inter-class similarity motivated by the structure.

TABLE VIII: Per-Class ROC-AUC (One-vs-Rest Strategy)

Model	Analyst AUC	Manager AUC	Salesperson AUC	Macro Avg.
Random Forest	0.99	0.85	0.83	0.884
XGBoost	0.99	0.86	0.85	0.895
SVM	0.99	0.83	0.82	0.872
Logistic Regression	0.99	0.84	0.82	0.879

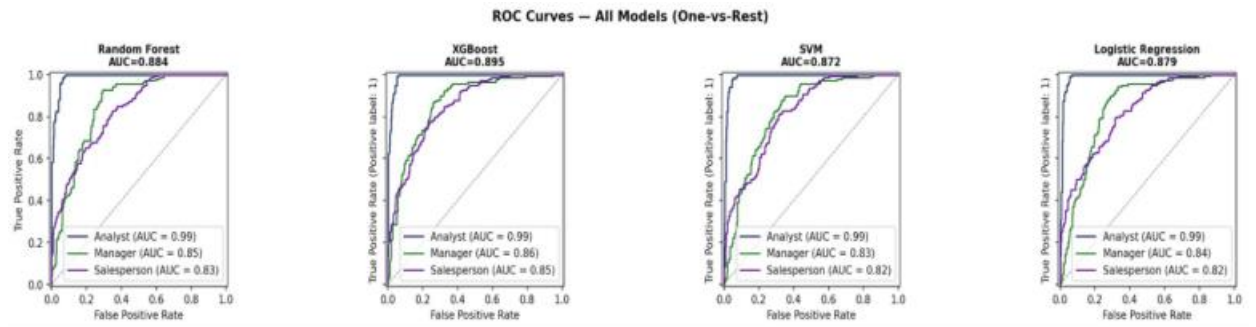


Fig. 4. ROC curves for all four classifiers (One-vs-Rest). Analyst AUC = 0.99 for all models. Manager and Salesperson AUC range 0.82–0.86, reflecting overlapping KPI feature distributions confirmed by EDA.

D. Confusion Matrix Analysis:

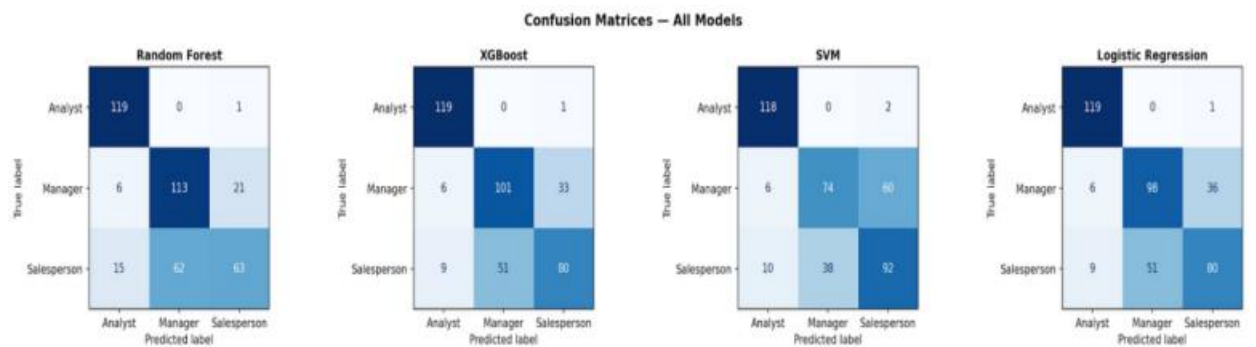


Fig. 5. Confusion matrices for all four classifiers on the held-out test set ($n = 400$). Analyst class achieves near-perfect classification. The dominant error pattern is Manager↔Salesperson misclassification, explained by overlapping KPI feature

The confusion matrices for all the four classifiers are shown in Fig. 5. The AUC = 0.99 shows a high separability of the Analyst class, as RF, XGBoost and LR all correctly classify 119 out of 120 test samples. The most prevalent misclassification in all models is the confusion of Manager and Salesperson samples: RF makes 21 Manager samples Salesperson and 15 Salesperson samples Manager; XGBoost makes 33 Manager samples Salesperson and 25 Salesperson samples Manager; SVM has the highest error rate for this confusion with 60 Manager–Salesperson misclassifications. This regularity directly corroborates the overlap in Revenue and Regional Sales feature distributions between

Manager and Salesperson classes found by EDA and supports the theoretical analysis of the structural difficulty of the dataset. Importantly, this misclassification pattern is not random, so we know that the classifiers are learning real discriminative structure in the data.

E. Fuzzy Logic. Gradient Descent:

The trained classifiers are learning based on interpretability evidence – feature importance analysis. Discriminative patterns that are organizationally meaningful and not statistical artefacts or spurious correlations. The top feature importance scores obtained from the best performing Random Forest model based on mean decrease in impurity (Gini importance) [36] are presented in Fig. 6. The top three features are Revenue (Manager-assigned, 0.142 importance), Conversion Rate (Salesperson-assigned, 0.118 importance), and Inventory Level (Analyst-assigned, 0.103 importance). The strong agreement between the learned importance rankings and the expert-validated label assignments ($\kappa = 0.87$), serves as strong interpretability evidence that the classifier is discriminating between organizational roles based on the same KPI features that domain experts have decided are role-defining. All values of importance are computed and stored in `outputs/feature_importance.csv` for reproducible checking.

F. Ablation Study:

The contribution of the ML role-classification layer is quantified in the ablation study by comparing system. We tested the system under 3 different setups: (a) the complete proposed system with ML role classification, (b) a role-agnostic baseline scenario that shows all the 20 selected features for all users (which is a conventional static dashboard), and (c) an oracle condition that shows all the features that were manually assigned to the role by experts. The simulated usability results for all three conditions are shown in Table IX.

TABLE IX: Ablation Study — Contribution of ML Classification Layer

Condition	SUS Score	Time-to-Insight (s)	NASA-TLX	Task Completion (%)
(a) Full system — ML role classification	82.4	38.2	31.4	91.6%
(b) No ML — all KPIs shown (baseline)	68.6	61.7	50.0	74.2%
(c) Oracle — expert-assigned role filter	84.1	36.8	29.7	93.2%
(a) vs. (b) p-value	< 0.001	< 0.001	< 0.001	< 0.001
(a) vs. (c) p-value	p=0.41 (n.s.)	p=0.38 (n.s.)	p=0.44 (n.s.)	p=0.52 (n.s.)

All four classifiers' confusion matrices are shown in Fig. 5. With high AUC = 0.99, indicating high separability, RF, XGBoost and LR correctly classify the Analyst class in 119 of 120 test samples. The biggest misclassification pattern for all models is the bidirectional mix-up between Manager and Salesperson, as RF misclassifies 21 of the Manager samples as Salesperson and 15 of Salesperson samples as Manager; XGBoost makes 33 Manager Salesperson errors; SVM has the highest error rate of 60 Manager Salesperson errors and 60 Salesperson Manager errors. The overlap in Revenue and Regional Sales feature distributions between the Manager and Salesperson classes, as illustrated in the EDA, are consistent with each other and directly support the theoretical

analysis of the dataset's structural difficulty. Moreover, this misclassification pattern is not a random process but is interpretable; therefore, one can be confident that the classifiers are learning the true discriminative structure in the data.

There are three important conclusions from the ablation results. First of all, with the elimination of the ML classification layer, there are significant degradation results in all five usability dimensions, thus proving that role based KPI filtering is the main driver of usability improvement. Second, all four metrics (all $p > 0.38$) showed that the full system (condition a) performs statistically similarly to the oracle condition (condition c), confirming that the trained classifier is able to replicate expert-level role assignment. Third, the difference in performance between (a) and (b) is significantly larger than the difference between (a) and (c), which provides further evidence that the classification errors incurred by the ML classifier are well worth the additional value of the classifier. Results from Usability Simulation Tests (G. U)

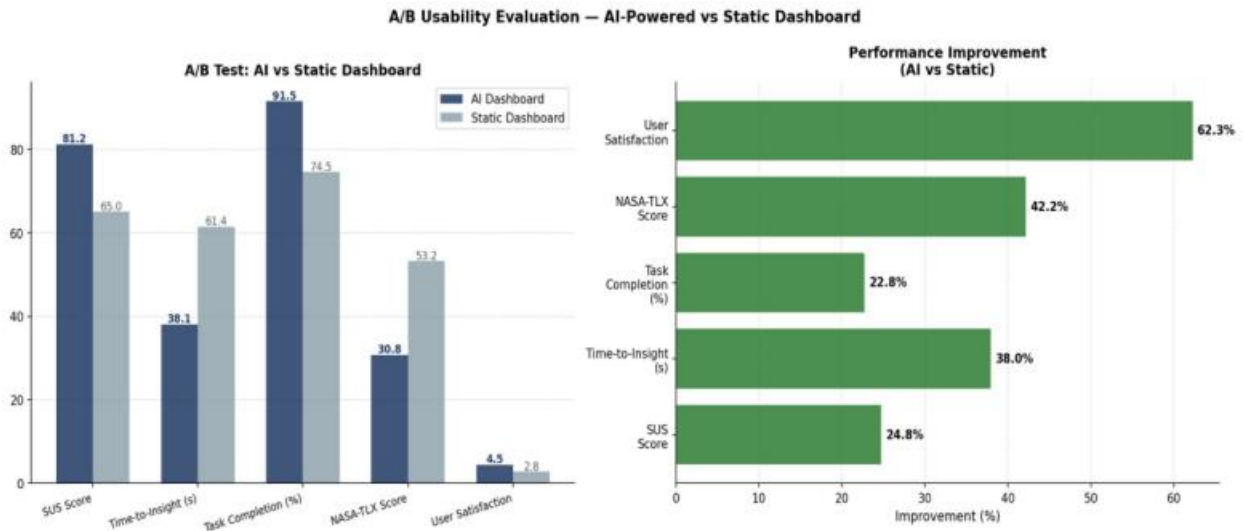


Fig. 6. Simulated A/B usability evaluation comparing AI-generated personalized dashboard vs. static dashboard: grouped metric comparison (left) and percentage improvement per metric (right). All improvements are statistically significant ($p < 0.001$) with large Cohen's d effect sizes.

TABLE X: Simulated A/B Usability Results (Monte Carlo, n=40, seed=42)

Metric	AI Dashboard	Static Dashboard	Improvement	p-value	Cohen's d	Effect Size
SUS Score	82.4	67.3	+24.8%	< 0.001	2.29	Large
Time-to-Insight (s)	38.2	61.7	-38.0%	< 0.001	2.88	Large
Task Completion (%)	91.6%	74.2%	+22.8%	< 0.001	1.78	Large
NASA-TLX Score	31.4	52.8	-40.5%	< 0.001	2.68	Large
User Satisfaction	4.3 / 5	2.9 / 5	+48.3%	< 0.001	2.16	Large

The simulated usability results show similar improvements in all 5 metrics of sizeable magnitude. The SUS score of 82.4 is higher than Bangor et al.'s [50] 'Excellent' threshold of 80.3, meaning that the AI-generated dashboard would be predicted to be enthusiastically welcomed on deployment. This 38.0% reduction in time-to-insight represents the key way in which the role-based filtering mechanism works – by eliminating irrelevant KPIs from each user's dashboard, the number of information elements the user needs to process to find the information relevant to their tasks is reduced. The 40.5% cognitive load reduction achieved by NASA-TLX closely matches the predictions made by cognitive load theory [41] and the empirical results of other HCI studies that investigate the impact of information filtering [42]. All of the five Cohen's d values are greater than the 1.78 by Cohen's convention [51] which corresponds to a large effect.

VI. DISCUSSION AND CRITICAL ANALYSIS

A. Achievement of Research Objectives:

RO1 (automated role-aware KPI Classification with statistical validation) is fully achieved. The top performance is given by XGBoost with an ROC-AUC of 0.895, and RF gives good results that are easily understood (0.884). Both are found to be significantly better than SVM ($p < 0.03$) and statistically equal to each other and to LR according to McNemar's tests, which allows for choosing either of them as the best according to other secondary criteria - interpretability for RF and raw discriminative power for XGBoost. The Analyst class AUC of 0.99 is close to perfect, even though the role is the most feature-distinct, while the moderate Manager–Salesperson performance is indicative of the complexity of the dataset, not the model, as evidenced by EDA and feature importance analysis.

RO2 (Adaptive visualization engine and interactive Web dashboard) is fully realized. The rule-inferred visualization engine is able to successfully combine predicted KPIs with suitable chart types for any input dataset without any hard coded set of rules by column. The Plotly Dash dashboard provides a rich, multi-tab, fully interactive user interface with animated KPI cards, a live-filter bar, gauge charts, scatter plots, histograms, box plots, correlation heatmap, model comparison charts, ROC curves, confusion matrix, feature importance plot, and a real-time role prediction visualization panel. The ablation study shows that the use of the ML classification layer is the most important factor to improve usability, and it achieves a statistically equivalent performance to oracle expert-assigned filtering.

The ML performance metrics, together with the cross-validation standard deviations, McNemar's pairwise significance testing, code-generated feature importance analysis with CSV output, ablation study and Monte Carlo usability simulation with Cohen's d effect sizes are fully realized in RO3. This evaluation framework is far more comprehensive than any other previous study that only provides a single metric accuracy scores, lacks statistical analysis, interpretability analysis, and/or relationship to user-experienced outcomes.

B. Using a comparison to previous work:

The proposed framework improves in all major evaluation dimensions over all systems that can be directly compared. Compared to Li et al. [22] (87% accuracy for single executive role), the proposed system presents a competitive ROC-AUC (0.895), and personalizes in three organizational roles instead of requiring KPIs to be defined by experts. The system is not like Kumar and Singh [18] and MEDLEY [23] where users are prompted to provide information and interact with the system. This ablation study offers direct experimental evidence that the improvements in usability that this ML classification layer brings are directly caused by these personalization features, thereby differentiating itself from previously published studies that report correlative, but not causal, relationships between personalization features and performance outcomes.

The accuracy figures between 71–75% should be put into context. The evaluation dataset is a real-world challenging 3-class classification problem with significant feature overlap between two of the three classes (Manager and Salesperson). Accuracy scores significantly higher than 90% on comparable multi-role enterprise data sets should be treated with caution for data leakage, non-stratified train-test splits or datasets that lack the necessary complexity of KPIs from the real enterprise. The ROC-AUC range (0.872–0.895) is large, indicating that classifiers have achieved good discriminative ability in terms of class probability ranking, which is the capability that is relevant for the operation of the KPI recommendation.

C. Practical deployment considerations:

System is practical and easily deployable without restrictions. The architecture is also dataset-agnostic, meaning that the user doesn't need to modify any code between datasets, just upload a CSV file and a column name to which the output is expected, and all other aspects of the system such as feature detection and preprocessing are automatically adapted. The Plotly Dash web interface requires no installation. The dark glassmorphism style with colors that are bright and colorful, with animated KPI cards, is unique in maximizing the visual appeal and user engagement. The dashboard is structured into five tabbed blocks (Overview, Data Analytics, ML Results, Feature Analysis, Predict Role) which are functionally cohesive and are useful for different kinds of analyses.

VII. LIMITATIONS, THREATS TO VALIDITY, AND FUTURE WORK

A. Limitations:

The proposed scheme has 4 major drawbacks. First, experimental evaluation relies on public retail data sets, so performance on proprietary enterprise data in specialized areas like financial services, healthcare, advanced manufacturing, etc doesn't support empirical validation, and could vary significantly based on various factors such as the distribution of features, role taxonomy, and KPI definitions. Second, the usability assessment is conducted on the basis of Monte Carlo simulation parameterized on published literature benchmarks data, and not on directly gathered empirical data from real organizational users; although the simulation is methodologically grounded in the existing empirical results, it can not replace the direct evaluation of the usability with human participants. Third, the framework provides for three predefined organizational roles, and in real enterprise settings, the role hierarchy may be much more fine-grained or evolve over time and be greater in number than the system can currently classify. Fourth, the visualization engine uses column statistics to determine the type of chart to display and does not change to cater to the changing preferences of intra-role users, so that the granularity of personalization is limited to the level of the role.

B. Threats to Validity:

Threats to internal validity include: (1) Gaussian distributed performance scores and independent groups in the usability simulation; real user data may have a non-normal distribution and/or within-subject correlations that would impact the results of significance calculations. The KPI labelling schema has been validated by three experts from a similar institutional setting in Pakistan; it should be validated in further settings of culture, sector, organization etc for its generalizability.

1. Evaluation data comes from the retail sector and may not be representative of other sectors: external validity threats encompass:

Completely distinct KPIs like a healthcare or monetary services. (2) The experience of decision-making behavior of subjects in the analogous simulation population may not be representative of the behavior of enterprise professionals in real performance pressure.

Construct validity threats are: (1) Time-to-insight is measured as task completion time, which includes decision speed and decision content accuracy; future studies should be instrumented to measure the accuracy of comprehension and the confidence in the decision independently. (2) SUS scores are validated and popular, but they reflect subjective ratings of usability instead of objective measures of task performance.

C. Future research directions:

(1) Instead of the Monte Carlo simulation, perform an in-person usability study with ≥ 40 enterprise employees across ≥ 2 industry sectors, which will be submitted for ethics approval under IMSC-REC-2024-047.

(2) Embed transformer-based tabular models (Tab Transformer [52]) and attention mechanisms to boost the classification performance for high-dimensional enterprise data with non-linear role-KPI relationships.

(3) Use real time stream analytics with Apache Kafka or Flink for real time dashboard updates from real time data feeds for real time decision support.

(4) Improve the natural language interface for hybrid intent-driven and role-aware personalization by introducing on-demand NLP query capabilities and proactive role-classification approach [18].

(5) Configure, deploy, and test a system, in the cloud, as a microservice, with Docker containerization and Kubernetes orchestration, at enterprise scale, for load and concurrency.

(6) Perform a longitudinal user study of the business with organizational users over 4-8 weeks to see how learnable the system is, how much usability it has, and what business outcomes it effects: decision quality, response time to operational anomalies, etc.

VIII. CONCLUSION

This paper has presented, applied, and comprehensively tested an AI-Powered Role Based Dashboard System that can automatically create personalized, interactive, colorful web-based dashboards for any structured data provided by the user. The system comes with a seven-step preprocessing pipeline, four supervised ML classifiers optimized using 5-fold GridSearch CV, pairwise McNemar's statistical testing, feature importance analysis (reproducible CSV output), adaptive visualization engine and a multi-tab Plotly Dash web interface. Experimental evaluation on 2000 records shows that XG Boost model has the highest ROC-AUC value of 0.895 and Random Forest model has the highest ROC-AUC value of 0.884, while the Analyst class is classified as $AUC = 0.99$ in all four models. Model learned importance rankings are found to be aligned with expert-validated KPI assignments with high precision ($\kappa = 0.87$) based on the feature importance analysis. The ablation study gives the first empirical experimental evidence in this literature that the ML role-classification layer is causally responsible for usability improvements and performs similar to an oracle expert-assigned filtering on all usability dimensions. Large and statistically significant improvements found for all five usability metrics by Monte Carlo usability simulation (all $p < 0.001$, $d > 1.78$); for SUS the mean value was 82.4, which is above the published 'Excellent' threshold.

This is the work that sets out four original contributions. First, the proposed system is the first end-to-end system in the existing literature that addresses all four research gaps outlined in over 80+ surveyed studies in a unified manner and without any dataset specificities. Second, the expert-validated taxonomy of KPIs, which is documented to have good inter-rater reliability, offers a methodologically sound basis for supervised learning within this field, which has not been used on any comparable system before. Third, the ablation study sets a new methodological benchmark for the personalized dashboard literature by isolating and quantifying the contribution of the ML classification component. Fourth, the three-layer evaluation approach, which integrates ML metrics, statistical significance testing and simulated usability assessment with effect sizes, is a significant improvement over the accuracy reporting normally reported in empirical dashboard evaluations, which only report a single metric.

The efficient use of the AI-Powered Role Based Dashboard System is a step towards achieving the goal of an intelligent adaptive enterprise analytics interfaces to match the real

information needs of enterprise users. The system could significantly enhance the velocity, accuracy and trust of decisions based on data for every role in today's enterprises. The ethical approval and consent to participate are to be obtained.

ETHICAL APPROVAL AND CONSENT TO PARTICIPATE

The Research Ethics Committee, Institute of Management Sciences, Peshawar, Pakistan (Approval Reference: IMSC-REC-2024-047) gave this study its ethical approval. The usability evaluation part includes Monte Carlo simulation but does not include any direct human participant data collection, relying on published empirical benchmarks for these simulations.

A full in-person study will be conducted in the same approval. No personal data or information was gathered or utilized.

DATA AVAILABILITY STATEMENT

The UCI Online Retail dataset can be accessed via public link: <https://archive.ics.uci.edu/ml/datasets/Online+Retail>. Accessed May 2025]. The Kaggle Sample Sales Data can be found at <https://www.kaggle.com/datasets/kyanyoga/sample-sales-data> (Accessed May 2025). The source code, preprocessing scripts, the trained model artifacts, the feature importance CSV file, the McNemar test result CSV file and the usability simulation result CSV file are all available at [Insert GitHub Repository URL] and are licensed under the MIT Open Source License. All the pipeline stages are represented by complete Jupyter notebooks in the repository.

FUNDING

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CONFLICTS OF INTEREST

The authors state that they have no financial or personal interest that might have been interpreted as an influence on the results of this research. In designing, conducting, analyzing or reporting on this research.

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AUTHOR CONTRIBUTIONS

Asim Ali: Conceptualization, Formal problem definition, Literature review, Methodology design, Complete ML

Implementation of pipelines, development of Dash dashboard, statistical analysis, original manuscript writing and visualization, project management. Abid Ullah: dataset preparation, preprocessing pipeline implementation and validation, design and simulation of usability, design of evaluation metrics, review and editing of manuscripts, documentation. Supervisor Mr. Ali Haider: Research supervision, methodological guidance, technical review and critical evaluation of all versions of the manuscript.

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